

Raphaël URFIN

Centre de Sciences des Données, 45 rue d'Ulm – 75005 Paris – France

About me

I am a first year PhD student at École Normale Supérieure - PSL in Paris supervised by Giulio Biroli (ENS) and Marc Mézard (Bocconi University) working on the theory of diffusion models. My background is in theoretical physics.

Education

- **PhD in Physics** **ENS-PSL, Paris**
Subject: Generative Diffusion and Statistical Physics
LPENS & Centre de Sciences des Données
Supervisors: Giulio Biroli (ENS) and Marc Mézard (Bocconi University)
2025 - 2028(exp.)
- **ENS Diploma** **ENS-PSL, Paris**
Interdisciplinary Diploma validating my studies at ENS.
2021-2025
- **M2 ICFP Theoretical Physics Track** **ENS-PSL, Paris**
First Class Honors, 17.38/20
2023 - 2024
Relevant classes: Advanced Statistical Physics, Disordered Systems, Machine Learning.
- **M1 ICFP** **ENS-PSL, Paris**
First Class Honors, 17.8/20
2022 - 2023
Relevant classes: Phase Transitions, Introduction to Quantum Field Theory, General Relativity.
- **Bachelor in Physics and Mathematics** **ENS-PSL, Paris**
First Class Honors, 17.38/20 (Physics) and 16.36/20 (Mathematics)
2021 - 2022

Publications

* denotes equal contribution.

- Tony Bonnaire*, **Raphaël Urfin***, Giulio Biroli, Marc Mézard.
Why Diffusion Models Don't Memorize: The Role of Implicit Dynamical Regularization in Training.
NeurIPS 2025, **Best Paper Award (4 papers among +20000 submissions)**
<https://openreview.net/forum?id=BSZqpqqqM0>

Conferences, Workshops and Seminars

Invited Talks.....

- **LemanTH 2026** **Lausanne, Switzerland**
Workshop
April 2026
 - Talk on the Memorization/Generalization transition in Diffusion Models.
- **Vitelli's Group Meeting** **Online**
Group meeting
March 2026
 - Talk on the non-reciprocal p-spins.
- **Scattering Club** **Online**
Invited Seminar
February 2026
 - 1h invited talk on the Memorization/Generalization transition in diffusion models. Webinar organized by the data science for astrophysics group of LPENS.

- **Voxel51 Best of NeurIPS 2025** **Online**
Invited Seminar *January 2026*
 - 30 min invited talk on the Memorization/Generalization transition in diffusion models.
- **Centre de Sciences des Données Internal Seminar** **Paris, France**
Invited Seminar *December 2025*
 - 1h invited talk on the Memorization/Generalization transition in diffusion models.
- **Machine Learning & Signal Processing @ ENS Lyon** **Lyon, France**
Invited Seminar *November 2025*
 - 1 hour invited talk on the Memorization/Generalization transition in diffusion models.

Contributed Talks.....

- **Journées de Physique Statistique 2026** **Paris, France**
Conference *January 2026*
 - 4-minute flash talk on the Memorization/Generalization transition in Diffusion Models.
- **EurIPS 2025** **Copenhagen, Denmark**
Conference *December 2025*
 - Poster+Oral Presentation of our paper accepted at NeurIPS 2025.
- **NeurIPS in Paris 2025** **Paris, France**
Workshop *November 2025*
 - Poster+Oral Presentation of our paper accepted at NeurIPS 2025.
- **StatPhys 29** **Florence, Italy**
Conference *July 2025*
 - 15-minute contributed talk on the Memorization/Generalization transition in diffusion models.
- **Youth in High Dimensions** **Trieste, Italy**
Workshop *July 2025*
 - 15-minute contributed talk on the Memorization/Generalization transition in diffusion models.
- **Journées de Physique Statistique 2025** **Paris, France**
Conference *January 2025*
 - 4-minute flash talk presenting results from my M2 internship.

Summer Schools.....

- **Beg Rohu Summer School of Statistical Physics** **Beg Rohu, France**
Summer School *June 2025 & June 2026*
 - **2026 Edition (Optimization and Control of Nonequilibrium Systems):** Lectures by international researchers (e.g. G. Peyré, G. Rotskoff ...). Presented an updated poster on Memorization/Generalization in diffusion models.
 - **2025 Edition (Learning with Machines, Physics and Mind):** Lectures on Machine Learning and Statistical Physics by international researchers (e.g., Yann LeCun, Julia Kempe, Stéphane Mallat, Marc Mézard ...). Presented initial poster on Memorization/Generalization in diffusion models.
- **Complex and Glassy Systems** **Cargese, France**
Summer School *July 2024*
 - Lectures on Statistical Physics and interdisciplinary applications by international researchers (e.g. Marc Mézard, Eric Vanden-Eijnden, Valentina Ros, Guy Bunin).

Awards and Grants

- 2025: Recipient of one of the four best paper awards at NeurIPS 2025. **(4 papers among +20000 submissions)**
- 2025: CDSN, Competitive PhD scholarship for ENS students

Reviewing

- Machine Learning Workshops: EurIPS 2025 Workshop Principles of Generative Modeling (PriGM), ICLR 2026 Workshop DeLTa, ICLR 2026 Workshop Sci4DL.

- Machine Learning Journals: TMLR

Skills

- Languages: French (Native), English (Fluent), Italian (Fluent).
- Software skills: Python (Pytorch), Matlab, Latex.

Teaching Experience

- TA Machine Learning (M2, Master ICFP, ENS-PSL), 11 hours, taught in English with Marc Lelarge and Leonardo Defilippis. Spring 2026.
- TA Mathematics Tutoring for first-year students in Physics at ENS, 36 hours, taught in French with Amir-Kian Kashani-Poor. Fall 2025.
- TA Stochastic Processes for Physics (M1 level, Master ICFP, ENS-PSL), 16 hours, taught in English with Marylou Gabrié. Fall 2025.
- "Khôlles" (Preparation for the oral exams of French "Grandes Écoles") in Mathematics and Physics for Classes Préparatoires PC, Lycée Stanislas (2022–2024).

Relevant Experience

- **Statistical Physics and Diffusion Models** **Bocconi University, Milano, Italy**
Research Internship *February 2025-May 2025*
 - Supervisor: Marc Mézard, Departement of Computing Sciences.
 - Subject: The Memorization/Generalization transition in diffusion models.
 - Resulted in a publication accepted at Neurips 2025
- **Statistical Physics and Diffusion Models** **École Normale Supérieure-PSL, France**
Research Internship *April 2024-January 2025*
 - Supervisor: Giulio Biroli, Centre de Sciences des Données.
 - Subject: The effect of implicit regularization in diffusion models.
- **Emergent behaviors in large ecosystems** **University of Cambridge, UK**
Research Internship *January-July 2023*
 - Supervisor: Camille Scalliet, Soft Matter Group, Department of Applied Mathematics and Theoretical Physics.
 - Using tools from disordered systems (Cavity Method, Random Matrix Theory...) to understand emergent collective behaviors in ecological systems.